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Case 33-2025: A 27-Year-Old Man with Abnormal Behaviors, Confusion, and Seizure

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CME

PRESENTATION OF CASE

Dr. Paavani S. Reddy (Psychiatry): A 27-year-old man was admitted to the medical service of this hospital because of abnormal behaviors and confusion.

The patient had been in his usual state of health until 2 months before the current presentation, when his family and outpatient mental health providers noticed abnormal behaviors and an increasing amount of paranoia and confusion after he had independently discontinued his medications prescribed to treat bipolar I disorder. The patient was noted to have repeatedly called his mother; he reported being persecuted, and his mother stated that he had been speaking in nonsensical, cryptic statements. The patient had no recollection of their conversation shortly after each phone call. On the day of admission, the patient became agitated upon trying to sleep, asking for the noise generated by the crickets outside to be silenced, and his partner brought him to the emergency department of this hospital for evaluation.

On evaluation in the emergency department, the patient's heart rate was 116 beats per minute, and all other vital signs were normal. A mental status examination showed no evidence of delusions or response to internal stimuli but was notable for a disorganized and tangential thought process with impaired orientation and attention, as well as limited short- and long-term recall. Lactated Ringer's solution was administered intravenously, and treatment with oral lorazepam was started. He was transferred to the acute psychiatric service of this hospital, and treatment with risperidone was initiated.

On the evening of the second hospital day, the patient had a tonic–clonic seizure that was witnessed by hospital staff members. Treatment with intravenous midazolam was administered, and the convulsions resolved. The patient was transferred to the medical service of this hospital.

On arrival, the patient was somnolent and sedated but reported no headache or weakness. A review of systems was negative for weight loss, palpitations, changes in stool, changes in vision, tremors, nausea, or previous seizurelike activity.

On examination, the oral temperature was 36.1°C, the blood pressure 144/70 mm Hg, the heart rate 129 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 99% while the patient was breathing ambient air. He appeared restless but not diaphoretic. No mydriasis, ophthalmoparesis, or proptosis was present. The thyroid was palpable, without nodules or enlargement. Cardiac auscultation was notable only for tachycardia, without murmur. The neurologic examination showed an unsteady gait; however, no motor weakness, cranial nerve asymmetry, pronator drift, dysmetria, or tremor was present.

The patient's psychiatric history was notable for bipolar I disorder and attention deficit–hyperactivity disorder. He had had multiple previous hospital admissions for mania and self-injurious behavior. His medical history was notable for remote hyperthyroidism, which had been treated with nonselective beta-blockers and no antithyroid agents. He took no medications but had previously received prescriptions for lithium carbonate, propranolol, bupropion, clonazepam, buspirone, and gabapentin. He was unemployed and lived with his partner in a suburb in Massachusetts. The patient drank alcohol daily and used marijuana often but did not use stimulants or other recreational drugs. The patient's father had bipolar I disorder.

The blood levels of electrolytes, alanine aminotransferase (ALT), aspartate aminotransferase (AST), total bilirubin, and alkaline phosphatase were normal, as were results of tests of kidney function. The white-cell count was 12,600 per microliter (reference range, 4000 to 11,000), with a neutrophil predominance, and the hemoglobin level and platelet count were normal. Urinalysis revealed ketonuria and mild proteinuria. Serum levels of lithium were below the limit of detection. Blood and urine toxicology screenings were positive for benzodiazepines; the blood alcohol level was undetectable. Screening tests for human immunodeficiency virus infection and syphilis were negative. Levels of vitamin B_{12} and folic acid were normal. Other laboratory test results are shown in Table 1. An electrocardiogram showed sinus tachycardia with no ST-segment or T-wave abnormalities. Subsequently, an electroencephalogram (EEG) showed no epileptiform discharges but showed generalized overriding beta frequencies.

Dr. Pedram Heidari: Computed tomography (CT) of the head was performed without the administration of intravenous contrast material. No evidence of acute hemorrhage, territorial infarction, mass effect, or midline shift was present. The cerebral ventricular system and cisternal spaces appeared normal for the patient's age, with no evidence of hydrocephalus or abnormal extra-axial fluid collections. No calcifications, bone abnormalities, or parenchymal hyperdensities were identified (Fig. 1A).

Magnetic resonance imaging (MRI) of the head was performed after the administration of intravenous contrast material to further investigate for subtle parenchymal abnormalities, vascular issues, and demyelinating disease. The overall diagnostic quality of the MRI was limited by motion artifacts, but there was no evidence of acute or chronic infarction, venous sinus thrombosis, intracranial mass, or abnormal enhancement consistent with neoplasm, infection, or inflammation. No restricted diffusion was observed on diffusion-weighted imaging, and susceptibility-weighted sequences showed no evidence of microhemorrhages or abnormal signals. No abnormal enhancement or white-matter signals suggestive of active demyelination were present (Fig. 1B).

Dr. Reddy: A lumbar puncture was performed as part of the evaluation of new seizure in this patient. The cerebrospinal fluid (CSF) was clear, and the opening pressure was normal. CSF analysis showed normal levels of protein and glucose, with 1 total nucleated cell per microliter (reference range, 0 to 5).

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Nadia V. Quijije: This 27-year-old man with bipolar I disorder, a history of hyperthyroidism, and daily alcohol use presented with a 2-month history of confusion, abnormal behavior, paranoia, nonsensical speech, and short-term memory deficits. The physical examination in the emergency department was notable for disorganized and tangential thinking, impaired orientation and attention, limited short- and long-term recall, and tachycardia. On the second hospital day, the patient had a tonic–clonic seizure that was witnessed by emergency department personnel. The results of the initial laboratory evaluation, EEG, imaging studies, and CSF analysis were essentially normal

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, This Hospital†	On Initial Presentation, This Hospital
Blood		
Hemoglobin (g/dl)	13.5–17.5	15.1
Hematocrit (%)	41.0–53.0	47.0
White-cell count (per μl)	4500–11,000	12,600
Differential count (per μl)		
Neutrophils	1800–7700	11,080
Lymphocytes	1000–4800	980
Monocytes	200–1200	420
Eosinophils	0–900	40
Basophils	0–300	50
Immature granulocytes	0–100	40
Platelet count (per μl)	150,000–400,000	363,000
Sodium (mmol/liter)	135–145	143
Potassium (mmol/liter)	3.4–5.0	4.0
Chloride (mmol/liter)	98–108	105
Carbon dioxide (mmol/liter)	23–32	23
Urea nitrogen (mg/dl)	8–25	14
Creatinine (mg/dl)	0.60–1.50	0.87
Glucose (mg/dl)	70–110	148
Globulin (g/dl)	1.9–4.1	3.8
Albumin (g/dl)	3.3–5.0	4.8
Alanine aminotransferase (U/liter)	10–40	13
Aspartate aminotransferase (U/liter)	10–55	16
Alkaline phosphatase (U/liter)	45–115	132
Total bilirubin (mg/dl)	0–1	0.5
Cerebrospinal fluid		
Total protein (mg/dl)	5–55	24
Glucose (mg/dl)	50–75	85
Nucleated-cell count (per μl)	0–5	1
Differential count (%)		
Neutrophils	—	0
Lymphocytes	—	76
Monocytes	—	24
Red-cell count (per μl)	0–5	1
Urine		
Color	Yellow	Yellow
Clarity	Clear	Turbid
pH	5.0–9.0	5.5
Specific gravity	1.001–1.035	1.034
Glucose	Negative	Negative
Ketones	Negative	2+

Table 1. (Continued.)

Variable	Reference Range, Adults, This Hospital†	On Initial Presentation, This Hospital
Leukocyte esterase	Negative	Negative
Nitrite	Negative	Negative
Blood	Negative	Negative
Protein	Negative	1+
Red cells (per high-power field)	0–2	0–2
White cells (per high-power field)	<10	<10
Bacteria	None	None

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

except for ketonuria and mild proteinuria. The physical examination was notable for tachycardia and the development of hypertension on the second day of hospitalization. Other features of his medical history to consider include previous lithium therapy and a thyroid disorder for which he had received no treatment other than nonselective beta-blockers.

In an attempt to explain this patient's clinical presentation, I will integrate his psychiatric and medical background with the foreground of abnormal behaviors, confusion, new seizure, and ketonuria. To develop a differential diagnosis, I will ask three central questions: First, is this patient having a manic episode? Second, is this patient withdrawing from alcohol? Third, does this patient have thyrotoxicosis?

Bipolar I Disorder

Given the patient's history of bipolar I disorder, we must consider whether this constellation of findings is due to mania. According to the criteria from the *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition (DSM-5), patients with a manic episode with psychotic features must display at least three of the following symptoms (or four if the patient's mood is only irritable), resulting in marked functional impairment: distractibility, impulsivity, grandiosity, flight of ideas, increased activity or agitation, decreased need for sleep, and talkativeness. It is important to note that these phenomena cannot be attributed to substance use or another medical condition.[1]

This patient has a 2-month history of symptoms consistent with mania, including abnormal behaviors, a noticeable change in functioning from baseline, and nonsensical speech, which could reflect flight of ideas. According to the DSM-5 criteria, psychotic features of a manic episode from bipolar I disorder can include hallucinations, delusions, disordered thinking, and a lack of awareness of reality — many of which this patient had. Given his history of psychiatric hospitalizations, and a temporal correlation of symptoms that started after he had discontinued his medications, an affective decompensation of his underlying bipolar I disorder is certainly a possible diagnosis. However, the findings of confusion, impaired attention, short- and long-term recall deficits, seizure, and ketonuria would not be explained by a manic episode related to bipolar I disorder. Taking all these factors into consideration, I wonder whether a medical cause could better explain this patient's presentation.

Alcohol Withdrawal

We are told that this patient has a history of daily alcohol use, and upon arrival at the hospital, he was noted to have an elevated heart rate, for which he received a dose of lorazepam. Subsequently, a generalized tonic–clonic seizure developed within 48 hours after presentation, a critical time period when patients are at peak risk for seizures related to alcohol withdrawal due to a relative γ-aminobutyric acid (GABA) deficiency.[2] Alcohol withdrawal syndrome occurs in patients with consistent, heavy, and daily alcohol use and usually begins within 6 to 12 hours after the last intake of alcohol. Patients commonly present with anxiety, gastrointestinal symptoms, and autonomic manifestations such as diaphoresis, tachycardia, tremor, and hypertension. Complicated alcohol withdrawal is characterized by the development of seizures, perceptual disturbances, agitation, confusion, disorientation, and pyrexia.[3]

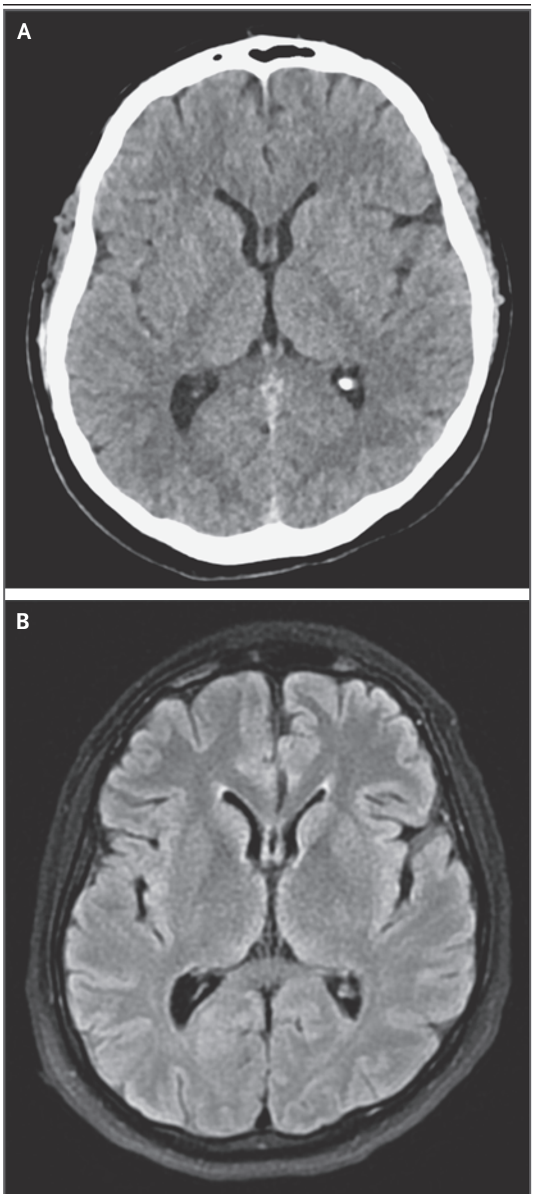


Figure 1. Imaging Studies of the Head.
CT of the head was performed without the administration of intravenous contrast material. A representative axial image (Panel A) shows normal brain parenchyma without evidence of hemorrhage, infarcts, or mass lesions. MRI of the head was performed after the administration of intravenous contrast material. A representative axial fluid-attenuated inversion recovery image (Panel B) shows no mass lesions, edema, or infarcts.

This patient certainly had signs of delirium and dysautonomia but had no tremor or diaphoresis. In addition, notably absent were the characteristic laboratory signs that are commonly associated with consistent, heavy, and daily consumption of alcohol. For example, in patients with alcohol-associated hepatitis, characteristic findings include a 2:1 ratio of AST to ALT, an elevated mean corpuscular volume from dyserythropoiesis, and thrombocytopenia secondary to myelosuppression.[4] Although this patient's ketonuria could be compatible with starvation and alcohol-associated ketoacidosis in the context of heavy use, complicated alcohol withdrawal would not explain the subacute change in behavior and the 2-month history of psychotic features that preceded the seizure.

Thyrotoxicosis

We are told that the patient has a history of hyperthyroidism and that aside from nonselective beta-blockers, he had received no treatment with antithyroid agents. Multiple features of this patient's case are consistent with a diagnosis of thyrotoxicosis, including the duration of his illness, the presence of maniform symptoms and cognitive impairment, persistent sinus tachycardia, and seizure (a rare finding). In addition, thyroid dysregulation, particularly hyperthyroidism, has been associated with ketonuria, which this patient has, although this is a nonspecific finding in the context of a protracted illness with unclear dietary intake.[5] Furthermore, the remainder of his diagnostic evaluation was largely unrevealing, including normal findings on MRI of the head, EEG, and CSF analysis — all of which would be expected in a patient with hyperthyroidism.[6] The most common cause of hyperthyroidism is Graves' disease[7]; however, chronic lithium exposure warrants consideration since it has a strong association with multiple diseases of the thyroid.

Graves' Disease

Graves' disease is an autoimmune disorder that results in autoantibody-mediated thyroidal activation, which leads to thyrotoxicosis. Neuropsychiatric symptoms of Graves' disease can include anxiety, depression, irritability, emotional lability, and cognitive dysfunction such as memory and concentration problems. More severe neuropsychiatric manifestations, including psychosis, delirium, and seizure, can develop in some patients, but these are rare and usually seen in those with severe thyrotoxicosis.[8] A physical examination may reveal a goiter and multiple ophthalmologic manifestations, including proptosis and diplopia.[9]

This patient did indeed present with subacute neuropsychiatric symptoms that could be consistent with Graves' disease, including cognitive dysfunction, maniform symptoms, and psychosis. However, psychosis and delirium are usually seen in advanced disease, which is unlikely in this patient given the absence of relevant physical examination findings. In addition, despite a diagnosis of hyperthyroidism, this patient had not received treatment with antithyroid medications, such as methimazole or propylthiouracil, which are standard treatments for Graves' disease. In thinking about subtypes of hyperthyroidism for which beta-adrenergic antagonists may be used as monotherapy to manage symptoms effectively, thyroiditis comes to mind.[10] Although there are various types of thyroiditis, given this patient's exposure history to lithium, I would like to specifically explore drug-induced thyroiditis.

LITHIUM-INDUCED THYROIDITIS

Lithium-induced thyroiditis typically manifests as a painless thyroiditis that is characterized by a triphasic course. First, there is a thyrotoxic phase, during which the thyroid tissue becomes inflamed and releases preformed thyroid hormones, which leads to transient hyperthyroidism. This phase can be asymptomatic or relatively mild and can last for weeks to months, with symptoms and signs including weight loss, tachycardia, heat intolerance, and tremor due to elevated thyroid hormone levels. Neuropsychiatric symptoms of the thyrotoxic phase of thyroiditis are similar to those seen with other causes of hyperthyroidism and include mood symptoms, cognitive changes, sleep disturbance, maniform symptoms, delirium, restlessness, and in rare cases, psychosis or seizures.[11,12] The thyrotoxic phase is usually followed by the more clinically apparent hypothyroid phase, which is seen once the gland is depleted of hormones or damaged. Most patients then enter the recovery phase, during which thyroid function typically returns to normal within 12 months.[13,14]

This patient's clinical presentation is consistent with drug-induced thyroiditis. Although lithium-induced thyroiditis usually occurs during lithium therapy, a delayed destructive thyroiditis can occur weeks to even months after treatment with lithium has been discontinued. Lithium accumulates and concentrates in the thyroid gland while patients are receiving therapy; therefore, the stored lithium in the thyroid tissue can continue to exert its damaging effects even after lithium therapy has been stopped.[15,16]

Diagnostic testing for lithium-induced thyroiditis usually involves thyroid-function assessments such as measuring the levels of thyrotropin, free thyroxine, and free triiodothyronine.[17] After confirmation of a thyrotoxic state, measurement of thyroid antibodies indicative of autoimmune thyroid disease, such as thyroid-stimulating immunoglobulin and thyrotropin-binding inhibitory immunoglobulin, can help distinguish between antibody-mediated thyroid diseases such as Graves' disease and drug-induced thyroiditis.[18,19] Moreover, imaging tests, such as thyroid scintigraphy with measurement of radioactive iodine uptake, can be very helpful in further determining the cause of hyperthyroidism by establishing or ruling out the presence of a diffusely active gland or an autonomous thyroid nodule (also known as a toxic nodule).[20]

In taking a step back to think about the diagnostic possibilities — bipolar I disorder, complicated alcohol withdrawal, and hyperthyroidism — it seems that a hyperthyroid state could best explain this patient's presentation, with lithium-induced thyroiditis and Graves' disease being at the top of my differential diagnosis.

CLINICAL IMPRESSION

Dr. Caitlin Adams: When we first evaluated this patient, our initial differential diagnosis included bipolar I disorder, owing to his previous diagnosis; however, his speech was nonpressured, and although his thoughts were disorganized, he had no flight of ideas. A separate thought disorder was also considered, given the recent onset of paranoia and concern for thought blocking, despite the fact that the patient reported no hallucinations when asked directly. Since the patient had a history of both alcohol and benzodiazepine use, withdrawal was closely considered, especially given his tachycardia and hypertension; however, no evidence of diaphoresis or a tremor was present on physical examination. Delirium was also suspected, since he was unable to answer questions related to orientation or attention, but he did not show signs of waxing and waning or distractibility, and he was not noted to have stimulus-bound behavior. Finally, we also considered the possibility of an overdose of bupropion, which can have stimulant effects and result in seizures, since the patient had previously received a prescription for this medication. However, his symptoms persisted during the hospital course, which made this an unlikely explanation.

Overall, the patient's presentation was unusual because it did not conform perfectly to any of the above diagnostic categories. He was not overtly manic, and yet, serial assessments resulted in consistent findings, which would be uncharacteristic of classic delirium. Although this episode could certainly represent decompensation of his underlying mood disorder, his history of hyperthyroidism was a potentially important clue, and we performed tests of thyroid function.

The patient had a thyrotropin level of less than 0.01 mIU per liter (reference range, 0.40 to 5.00), a total thyroxine level of 21.2 μg per deciliter (reference range, 4.5 to 10.9), and a free thyroxine level of 3.7 ng per deciliter (48 pmol per liter; reference range, 0.9 to 1.8 ng per deciliter [12 to 23 pmol per liter]), findings that are consistent with a diagnosis of thyrotoxicosis. From a psychiatric standpoint, the patient's symptoms were most consistent with a diagnosis of psychotic disorder due to another medical condition.[21] Diagnostic criteria include prominent delusions or hallucinations and laboratory evidence confirming that the behavioral disturbance is the direct consequence of another medical condition. In addition, the behavioral disturbance cannot be better explained by another psychiatric condition, does not occur exclusively during delirium, and results in significant distress or impairment — all features that were present in this patient.

CLINICAL DIAGNOSIS

Psychotic disorder due to another medical condition.

DR. NADIA V. QUIJIJE'S DIAGNOSIS

Thyrotoxicosis in the context of previous lithium exposure.

DISCUSSION OF MANAGEMENT

Dr. Michelle Rengarajan: The endocrinology service was consulted owing to the patient receiving a diagnosis of a psychotic disorder in the context of thyrotoxicosis. Thyrotoxicosis with a low level of thyrotropin can occur through autonomous thyroid hormone production, release of preformed hormone, or exogenous hormone administration. Thyroid autonomy can result from an autonomous thyroid nodule, in which a clonal population of cells in the thyroid produces hormone independently of thyrotropin, or from Graves' disease, in which autoantibodies directly stimulate the thyrotropin receptor and lead to thyrotropin-independent production of thyroid hormone throughout the thyroid gland. Alternatively, since the thyroid gland stores substantial amounts of thyroid hormone, the release of preformed thyroid hormone owing to a destructive thyroiditis, typically through direct damage to the thyroid tissue, leads to excess circulating thyroxine with consequent suppression of thyrotropin. Finally, exogenous levothyroxine is directly measured in thyroxine assays; if the dose of levothyroxine is too high, the level of thyroxine becomes elevated and leads to suppression of thyrotropin.

These possibilities can be distinguished from one another with a straightforward workup (Fig. 2). Graves' disease is associated with both increased secretion of intrathyroidal triiodothyronine and increased conversion of thyroxine to triiodothyronine[22]; therefore, a triiodothyronine:thyroxine ratio of greater than 20 (with triiodothyronine and thyroxine measured in nanograms per deciliter and micrograms per deciliter, respectively) is suggestive of Graves' disease. Elevated levels of antibodies targeting the thyrotropin receptor (thyroid-stimulating immunoglobulin or thyroid-binding inhibitory immunoglobulin) indicate Graves' disease.[23] Finally, thyroid scintigraphy with measurement of radioactive iodine uptake can identify the location of active production of thyroid hormone. When thyrotropin is undetectable, a high uptake of radioactive iodine is consistent with either an autonomous thyroid nodule (if uptake is localized)

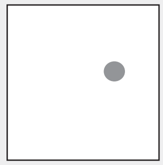
	Thyroid Autonomy from Graves' Disease	Thyroid Autonomy from Toxic Nodule	Exogenous Thyroid Hormone	Release of Preformed Hormone	This Patient
T3:T4 ratio	>20	Usually <20	<20	<20	**9.9**
Iodine-123 uptake and scan of the thyroid			No uptake	No uptake	**No uptake**
Thyroid-stimulating immunoglobulin	Present	Absent	Absent	Absent	**Absent**
Thyroid-binding inhibitory immunoglobulin	Present	Absent	Absent	Absent	**Absent**

Figure 2. Identifying the Cause of Hyperthyroidism.

The causes of hyperthyroidism can be distinguished from each other on the basis of the ratio of blood levels of total triiodothyronine (T3, measured in nanograms per deciliter) to thyroxine (T4, measured in micrograms per deciliter), the results of thyroid scintigraphy with measurement of radioactive iodine uptake, and the presence or absence of thyroid-stimulating immunoglobulin and thyrotropin-binding inhibitory immunoglobulin.

or Graves' disease (if uptake occurs throughout the gland). If thyrotoxicosis is caused by either release of preformed hormones owing to a destructive thyroiditis or exogenous administration of thyroid hormone, production of thyroid hormone is suppressed, and uptake of radioactive iodine is absent. Only a patient's clinical history can distinguish between destructive thyroiditis and exogenous administration of thyroid hormone. Although it is not essential to obtain a comprehensive ultrasound of the thyroid to determine the cause of thyrotoxicosis, it can be helpful if increased blood flow is present (a finding suggestive of Graves' disease) or if a nodule is visualized that corresponds to an area of increased uptake on a radioactive iodine scan.

Dr. Heidari: This patient underwent ultrasonography of the thyroid, which showed no nodules, cysts, or focal masses. The thyroid gland appeared to be homogeneous in echotexture and normal in size, with no evidence of structural abnormalities or excess blood flow on color or power Doppler imaging (Fig. 3).

Thyroid scintigraphy with iodine-123 showed diffuse and uniform suppression, with 1.2% uptake (Fig. 4A). In the context of clinical hyperthyroidism, the uniformly decreased uptake is consistent with an excess of thyroid hormone from hormone release or exogenous hormone use. This contrasts with typical findings in patients with Graves' disease (in whom an iodine-123 uptake scan would show uniformly increased uptake of iodine-123) and in those with an autonomous thyroid nodule (in whom a scan would show focal elevated uptake of iodine-123 with uniformly suppressed uptake in the remainder of the thyroid gland) (Fig. 4B and 4C).

Dr. Rengarajan: This patient had a triiodothyronine:thyroxine ratio of 9.9, negative tests for thyroid-stimulating immunoglobulin and thyroid-binding inhibitory immunoglobulin, and undetectable uptake of iodine-123 on thyroid scintigraphy, all findings consistent with either excess exogenous thyroid hormone or release of pre-

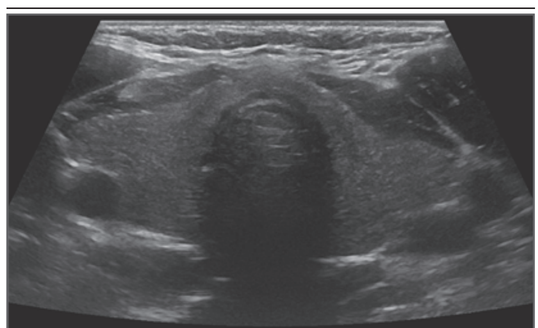


Figure 3. Thyroid Ultrasound Image.

A representative screen capture from transverse ultrasonography of the thyroid shows a homogeneous thyroid parenchyma with no defined thyroid nodules.

Figure 4. Radioactive Iodine Uptake and Scan of the Thyroid.

Thyroid scintigraphy with measurement of radioactive iodine uptake was performed with the use of iodine-123. Panel A shows uniformly decreased uptake of iodine-123 in the thyroid, which could be a result of thyroid hormone release or exogenous thyroid hormone consumption. Panel B shows uniformly increased tracer uptake throughout the thyroid gland and visualization of the thyroid pyramidal lobe (arrow), findings consistent with Graves' disease in the context of hyperthyroidism. Panel C shows a solitary hyperfunctioning autonomous thyroid nodule, also known as a toxic nodule (black arrow), in the right thyroid lobe with suppression of the surrounding thyroid tissue; a sternal notch marker (white arrow) is also shown.

formed thyroid hormone owing to a destructive thyroiditis. Although this patient's history is potentially unreliable, we do not have a clear indication of exogenous thyroid hormone administration, and he has a history of previous, self-limited hyperthyroidism. Taken together, these findings are most consistent with a diagnosis of subacute painless thyroiditis.

Did this patient's previous lithium use contribute to his current presentation? Typically, lithium exposure confers a high risk of goiter and hypothyroidism. In addition, hyperthyroidism, specifically painless subacute thyroiditis, occurs more often in patients with previous exposure to lithium.[24] It is unclear whether the risk of hyperthyroidism decreases over time after cessation of lithium therapy, and the underlying mechanism remains unknown. Therefore, although it is possible that his painless subacute thyroiditis is linked to lithium use, this is not certain.

We recommended that the patient continue treatment with beta-blockers, increasing the dose as allowed by his heart rate, to control the symptoms caused by hyperthyroidism. In patients with painless subacute thyroiditis, hyperthyroidism is expected to resolve as preformed thyroid hormone is depleted. Antithyroid agents, such as methimazole and propylthiouracil, interfere with thyrotropin-mediated thyroid hormone production. Given that preformed thyroid hormone drives the symptoms and that production of additional thyroid hormone is suppressed in painless thyroiditis, these agents do not have a role in treatment.

Upon depletion of circulating hormone, temporary hypothyroidism can develop, owing to either delayed recovery from thyrotropin suppression or an inability of the damaged thyroid tissue to produce adequate hormone levels. When hypothyroidism develops in patients with subacute painless thyroiditis, I typically start treatment with levothyroxine. After 6 months, I then stop treatment with levothyroxine and assess whether the patient has normal thyroid function.

FOLLOW-UP

Dr. Rengarajan: The patient's mentation and behavior improved over the next 5 days of hospitalization in the medical service, and no further seizure activity was witnessed. We advised close follow-up in the outpatient endocrine and psychiatric clinics to monitor the results of thyroid-function tests over time and to evaluate him for impending hypothyroidism, as well as to treat his underlying mood disorder. Unfortunately, the patient was lost to follow-up after discharge.

FINAL DIAGNOSIS

Subacute painless thyroiditis.

This case was presented at Psychiatry Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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